

$$\begin{aligned} & \rightarrow \frac{1}{16\pi^2} \left(-\frac{1}{3} g_2^4 \text{LF}_{3,0} [m_2] - \frac{1}{2} g_2^4 \text{LF}_{4,-1} [m_2] + \frac{8}{15} g_2^4 \text{LF}_{5,-2} [m_2] + \right. \\ & \frac{1}{24} g_2^2 \left(6 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} - 7 \sum_{\text{p}} g_2^2 \right) \text{LF}_{3,0} [m_{\text{l}}^{\text{p}}] + \frac{1}{16} \left(-4 g_2^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} + 7 \sum_{\text{p}} g_2^4 \right) \text{LF}_{4,-1} [m_{\text{l}}^{\text{p}}] - \\ & \frac{2}{15} \sum_{\text{p}} g_2^4 \text{LF}_{5,-2} [m_{\text{l}}^{\text{p}}] + \frac{1}{8} g_2^2 \left(6 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} + 6 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} - 7 \sum_{\text{p}} g_2^2 \right) \text{LF}_{3,0} [m_{\text{q}}^{\text{p}}] + \\ & \frac{3}{16} g_2^2 \left(-4 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} - 4 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} + 7 \sum_{\text{p}} g_2^2 \right) \text{LF}_{4,-1} [m_{\text{q}}^{\text{p}}] - \frac{2}{5} \sum_{\text{p}} g_2^4 \text{LF}_{5,-2} [m_{\text{q}}^{\text{p}}] - \frac{1}{6} g_2^4 \text{LF}_{3,0} [\tilde{\mu}] - \\ & \frac{1}{4} g_2^4 \text{LF}_{4,-1} [\tilde{\mu}] + \frac{4}{15} g_2^4 \text{LF}_{5,-2} [\tilde{\mu}] + \frac{3}{2} g_1^4 \tilde{\mu}^2 \text{LF}_{3,2,-1} [m_1, \tilde{\mu}] - g_1^4 \tilde{\mu}^2 \text{LF}_{3,3,-2} [m_1, \tilde{\mu}] - \\ & g_1^4 \tilde{\mu}^2 \text{LF}_{4,2,-2} [m_1, \tilde{\mu}] + 2 g_2^4 \text{LF}_{2,1,0} [m_2, \tilde{\mu}] + \frac{3}{4} g_2^4 \text{LF}_{2,2,-1} [m_2, \tilde{\mu}] + g_2^4 m_2^2 \text{LF}_{2,2,0} [m_2, \tilde{\mu}] - \\ & g_2^4 \text{LF}_{3,1,-1} [m_2, \tilde{\mu}] + g_2^4 \text{LF}_{3,2,-2} [m_2, \tilde{\mu}] - \frac{3}{2} g_2^4 (m_2^2 - 3 \tilde{\mu}^2) \text{LF}_{3,2,-1} [m_2, \tilde{\mu}] - \\ & g_2^4 \text{LF}_{3,3,-3} [m_2, \tilde{\mu}] + g_2^4 (m_2^2 - 3 \tilde{\mu}^2) \text{LF}_{3,3,-2} [m_2, \tilde{\mu}] - g_2^4 \text{LF}_{4,2,-3} [m_2, \tilde{\mu}] + \\ & g_2^4 (m_2^2 - 3 \tilde{\mu}^2) \text{LF}_{4,2,-2} [m_2, \tilde{\mu}] - \frac{1}{4} g_2^2 (\overline{a_e^{\text{pr}}} a_e^{\text{pr}} + 2 \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}}) \text{LF}_{4,1,-1} [m_{\text{l}}^{\text{p}}, m_{\text{e}}^{\text{r}}] + \\ & \frac{1}{2} g_2^2 (\overline{a_e^{\text{pr}}} a_e^{\text{pr}} + \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}}) \text{LF}_{5,1,-2} [m_{\text{l}}^{\text{p}}, m_{\text{e}}^{\text{r}}] - \\ & \frac{3}{4} g_2^2 (\overline{a_d^{\text{pr}}} a_d^{\text{pr}} + 2 \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}}) \text{LF}_{4,1,-1} [m_{\text{q}}^{\text{p}}, m_{\text{d}}^{\text{r}}] + \\ & \frac{3}{2} g_2^2 (\overline{a_d^{\text{pr}}} a_d^{\text{pr}} + \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}}) \text{LF}_{5,1,-2} [m_{\text{q}}^{\text{p}}, m_{\text{d}}^{\text{r}}] - \frac{3}{4} \overline{y_d^{\text{pr}}} y_d^{\text{sr}} \overline{y_u^{\text{st}}} y_u^{\text{pt}} \text{LF}_{2,1,0} [m_{\text{q}}^{\text{p}}, m_{\text{q}}^{\text{s}}] + \\ & \frac{3}{4} \overline{y_d^{\text{pr}}} y_d^{\text{sr}} \overline{y_u^{\text{st}}} y_u^{\text{pt}} \text{LF}_{3,1,-1} [m_{\text{q}}^{\text{p}}, m_{\text{q}}^{\text{s}}] - \frac{3}{4} g_2^2 (\overline{a_u^{\text{pr}}} a_u^{\text{pr}} + \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}}) \text{LF}_{2,2,0} [m_{\text{q}}^{\text{p}}, m_{\text{u}}^{\text{r}}] + \\ & \frac{3}{4} g_2^2 (\overline{a_u^{\text{pr}}} a_u^{\text{pr}} - \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}}) \text{LF}_{3,1,0} [m_{\text{q}}^{\text{p}}, m_{\text{u}}^{\text{r}}] + \frac{3}{4} g_2^2 \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} \text{LF}_{3,2,-1} [m_{\text{q}}^{\text{p}}, m_{\text{u}}^{\text{r}}] + \\ & \frac{3}{4} g_2^2 (-\overline{a_u^{\text{pr}}} a_u^{\text{pr}} + \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}}) \text{LF}_{4,1,-1} [m_{\text{q}}^{\text{p}}, m_{\text{u}}^{\text{r}}] - \frac{3}{4} \overline{y_d^{\text{pr}}} y_d^{\text{sr}} \overline{y_u^{\text{st}}} y_u^{\text{pt}} \text{LF}_{2,1,0} [m_{\text{q}}^{\text{s}}, m_{\text{q}}^{\text{p}}] + \\ & \frac{3}{4} \overline{y_d^{\text{pr}}} y_d^{\text{sr}} \overline{y_u^{\text{st}}} y_u^{\text{pt}} \text{LF}_{3,1,-1} [m_{\text{q}}^{\text{s}}, m_{\text{q}}^{\text{p}}] + \frac{3}{4} g_2^2 (\overline{a_u^{\text{pr}}} a_u^{\text{pr}} + \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}}) \text{LF}_{3,1,0} [m_{\text{u}}^{\text{r}}, m_{\text{q}}^{\text{p}}] + \\ & \frac{3}{4} g_2^2 (\overline{a_u^{\text{pr}}} a_u^{\text{pr}} + \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}}) \text{LF}_{3,2,-1} [m_{\text{u}}^{\text{r}}, m_{\text{q}}^{\text{p}}] - \\ & \frac{9}{4} g_2^2 (\overline{a_u^{\text{pr}}} a_u^{\text{pr}} + \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}}) \text{LF}_{4,1,-1} [m_{\text{u}}^{\text{r}}, m_{\text{q}}^{\text{p}}] + \\ & \frac{3}{4} g_2^2 (\overline{a_u^{\text{pr}}} a_u^{\text{pr}} + \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}}) \text{LF}_{5,1,-2} [m_{\text{u}}^{\text{r}}, m_{\text{q}}^{\text{p}}] - \frac{3}{2} \overline{y_d^{\text{pr}}} y_d^{\text{sr}} \overline{y_u^{\text{st}}} y_u^{\text{pt}} \text{LF}_{2,1,0} [m_{\text{u}}^{\text{t}}, m_{\text{d}}^{\text{r}}] + \\ & \frac{3}{2} \overline{y_d^{\text{pr}}} y_d^{\text{sr}} \overline{y_u^{\text{st}}} y_u^{\text{pt}} \text{LF}_{3,1,-1} [m_{\text{u}}^{\text{t}}, m_{\text{d}}^{\text{r}}] + \frac{1}{2} g_1^4 \tilde{\mu}^2 \text{LF}_{3,2,-1} [\tilde{\mu}, m_1] + \frac{5}{4} g_1^2 g_2^2 \text{LF}_{4,1,-2} [\tilde{\mu}, m_1] - \\ & \frac{1}{4} g_1^4 \tilde{\mu}^2 \text{LF}_{4,2,-2} [\tilde{\mu}, m_1] - g_1^2 g_2^2 \text{LF}_{5,1,-3} [\tilde{\mu}, m_1] - 4 g_2^4 \text{LF}_{3,1,-1} [\tilde{\mu}, m_2] - 2 g_2^4 \text{LF}_{3,2,-2} [\tilde{\mu}, m_2] + \\ & \frac{1}{2} g_2^4 (-2 m_2^2 + 3 \tilde{\mu}^2) \text{LF}_{3,2,-1} [\tilde{\mu}, m_2] + \frac{27}{4} g_2^4 \text{LF}_{4,1,-2} [\tilde{\mu}, m_2] + \frac{1}{2} g_2^4 \text{LF}_{4,2,-3} [\tilde{\mu}, m_2] + \\ & \frac{1}{4} g_2^4 (4 m_2^2 - 3 \tilde{\mu}^2) \text{LF}_{4,2,-2} [\tilde{\mu}, m_2] - 3 g_2^4 \text{LF}_{5,1,-3} [\tilde{\mu}, m_2] - 2 g_1^2 g_2^2 \text{LF}_{2,2,1,-2} [m_2, \tilde{\mu}, m_1] - \\ & \frac{3}{2} g_1^2 g_2^2 (m_1 m_2 + 2 \tilde{\mu}^2) \text{LF}_{2,2,1,-1} [m_2, \tilde{\mu}, m_1] + g_1^2 g_2^2 \text{LF}_{3,2,1,-3} [m_2, \tilde{\mu}, m_1] + \\ & g_1^2 g_2^2 (m_1 m_2 + 2 \tilde{\mu}^2) \text{LF}_{3,2,1,-2} [m_2, \tilde{\mu}, m_1] + \frac{1}{4} \tilde{\mu}^2 \overline{y_e^{\text{pr}}} \overline{y_e^{\text{st}}} y_e^{\text{pt}} y_e^{\text{sr}} \text{LF}_{2,2,1,-1} [m_{\text{l}}^{\text{p}}, m_{\text{l}}^{\text{s}}, m_{\text{e}}^{\text{r}}] + \\ & \frac{3}{4} (-\overline{y_u^{\text{st}}} y_u^{\text{pt}} \overline{a_d^{\text{pr}}} a_d^{\text{sr}} + \tilde{\mu}^2 \overline{y_d^{\text{pr}}} \overline{y_d^{\text{st}}} y_d^{\text{pt}} y_d^{\text{sr}}) \text{LF}_{2,2,1,-1} [m_{\text{q}}^{\text{p}}, m_{\text{q}}^{\text{s}}, m_{\text{d}}^{\text{r}}] + \\ & \frac{3}{4} (-\overline{y_d^{\text{st}}} y_d^{\text{pt}} \overline{a_u^{\text{pr}}} a_u^{\text{sr}} + \tilde{\mu}^2 \overline{y_u^{\text{pr}}} \overline{y_u^{\text{st}}} y_u^{\text{pt}} y_u^{\text{sr}}) \text{LF}_{2,1,1,0} [m_{\text{q}}^{\text{p}}, m_{\text{q}}^{\text{s}}, m_{\text{u}}^{\text{r}}] + \\ & \frac{3}{4} (\overline{y_d^{\text{st}}} y_d^{\text{pt}} \overline{a_u^{\text{pr}}} a_u^{\text{sr}} - \tilde{\mu}^2 \overline{y_u^{\text{pr}}} \overline{y_u^{\text{st}}} y_u^{\text{pt}} y_u^{\text{sr}}) \text{LF}_{$$